

Evaluating Spectral Interpretability in Brazilian Portuguese Audio Deepfakes via Causal Occlusion

Como as pistas acústicas de um detector XLS-R mudam entre zero-shot e adaptação leve — TODO:
confirmar/traduzir

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ABSTRACT

TODO: abstract.

KEYWORDS

audio deepfake detection, anti-spoofing, explainability, SHAP, spectral occlusion, MFCC, cross-lingual, wav2vec 2.0, XLS-R, Brazilian Portuguese

1 INTRODUCTION

TODO.

2 RELATED WORK

2.1 Self-Supervised Front-Ends for Anti-Spoofing

Tak et al. [4] TODO.

2.2 Explainability in Audio Deepfake Detection

Grinberg et al. [3] and Fullwood and Monroe [2] TODO.

2.3 Cross-Lingual Generalization and Real-World Deployment

Channing et al. [1] TODO.

3 METHODOLOGY

3.1 Detectors Under Study

TODO.

3.2 Pre-processing

TODO.

3.3 Acoustic Features (MFCC)

TODO.

3.4 Spine 1 — Surrogate + SHAP

TODO.

3.5 Spine 2 — Spectral Causal Occlusion

TODO.

3.6 Confirmatory Statistics

TODO.

Table 1: Experimental configuration.

Parameter	Value
Sample rate	16,000 Hz
Fixed window	64,600 samples (≈ 4.04 s)
Front-end	wav2vec 2.0 / XLS-R (frozen)
Checkpoint	nii-yamagishilab/mms-300m-anti-deepfake
Adaptation head (D_{ad})	logistic regression on 1024-d embeddings
Surrogate model	boosted trees $\rightarrow$ TreeSHAP
MFCC features	13 coeff. $\times$ (mean, std) = 26
Dataset	AKCIT-Deepfake/BRSpeech-DF
Train per class (D_{ad} head)	1,500 (balanced)
Analysis set per class (test)	1,500 (balanced)
Positive class	spoof
Seed	42

Table 2: Master table schema (one row per audio clip; $_{zs}$ = zero-shot, $_{ad}$ = adapted).

Column	Description
sample_id	unique clip identifier
ground_truth	0 = bonafide, 1 = spoof
spoof_algorithm	generator/attack (null if absent)
p_spoof_{zs,ad}	continuous $P(\text{spoof})$ per detector
pred_{zs,ad}	decision at EER threshold
quadrant_{zs,ad}	TP / TN / FP / FN
mfcc{i}_mean	mean of i -th MFCC ($i = 1..13$)
mfcc{i}_std	std of i -th MFCC ($i = 1..13$)

Table 3: Detection performance and surrogate fidelity (performance_table).

Detector	EER	MCC	Accuracy	Surrogate R^2
D_{zs} (zero-shot)	TODO	TODO	TODO	TODO
D_{ad} (adapted)	TODO	TODO	TODO	TODO

4 EXPERIMENTAL SETUP

5 RESULTS

5.1 Zero-Shot vs. Adapted Performance

TODO.

6 EXPLAINABILITY ANALYSIS

6.1 Acoustic Importance via SHAP

TODO.

6.2 Spectral Occlusion Profiles

TODO.

6.3 Confirmatory Tests

TODO.

7 DISCUSSION AND LIMITATIONS

TODO.

8 CONCLUSION AND FUTURE WORK

TODO.

REFERENCES

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